Practical No:02

Aim:
To insert single and multiple documents into a MongoDB collection.

Theory:

In MongoDB, data is stored in the form of **documents**, which are grouped inside **collections**. These documents follow the BSON (Binary JSON) format and contain key–value pairs similar to JSON objects. MongoDB provides powerful insertion methods to add data to collections. The two most commonly used methods are **insertOne()** and **insertMany()**, which allow inserting single and multiple documents respectively. Understanding how these functions work is fundamental for performing CRUD operations and managing data within a MongoDB database.

The **insertOne()** method is used when a single document needs to be added to a collection. It accepts one JSON-like document as a parameter. If the collection does not exist at the time of insertion, MongoDB automatically creates the collection before storing the document. Upon successful insertion, MongoDB generates a unique identifier, _id, for the new document. This _id serves as the primary key and helps in fast retrieval, indexing, and referencing of the document. Developers may also manually assign an _id, but in most cases, the auto-generated ObjectId is used.

The **insertMany()** method is used to insert multiple documents in a single call. It takes an array of documents as input. This method is more efficient than inserting documents one-by-one because it reduces the number of interactions with the database and enhances performance, especially in applications dealing with large datasets. Each document inserted through insertMany() also receives its own _id field. If any document insertion depends on certain validations or constraints, MongoDB provides options to control whether the entire batch fails or partial insertions are allowed.

Commands:

```
db.students.insertOne({ name: " Pratiksha ", dept: "CSE", age: 25 })
db.students.insertMany([ { name: "Pranjali", dept: "ETC", age: 22 }, { name: "Ishan", dept:
"ME", age: 27 } ])
```

Expected Output:

Documents are added with auto-generated _id.

Viva Questions:

Q1. What is the advantage of using insertMany() over multiple insertOne() calls?

insertMany() is more efficient because it reduces database communication and improves performance.

Q2. In which format does MongoDB store data internally?
MongoDB stores data in BSON (Binary JSON) format.

Q3. Which command is used to insert multiple documents at once?
The insertMany() command is used to insert multiple documents in a single operation.

Q4. Which command is used to insert a single document in MongoDB?
The insertOne() command is used to insert a single document into a collection.

Conclusion: In this practical, we successfully learned how to insert both single and multiple documents into a MongoDB collection using the insertOne() and insertMany() commands. We observed that MongoDB automatically creates a collection if it does not already exist and assigns a unique _id to each inserted document.